

NanoGT Match Report: Alpha-mannosidosis

Disease: Alpha-mannosidosis (ORPHA:61)

Primary gene: MAN2B1

Gene CDS: 3033 bp

Inheritance: Autosomal recessive

Target tissues: CNS, liver

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	ABO-101	AAV9	8.5/10	High	phase1/2
2	RGX-121	AAV9	8.5/10	High	phase3
3	ST-920	AAV2/6	7.7/10	High	phase1/2
4	DTX201	AAV8	6.8/10	Medium	phase2
5	Hemgenix	AAV5	6.7/10	Medium	approved

Match #1: ABO-101

Precedent disease: Mucopolysaccharidosis type IIIB

Vector: AAV9

Tissue target: CNS

Composite score: 8.5 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	1.0	2.0
Tissue tropism	2.0	2.0
Protein class	2.0	2.0
Pathway similarity	2.0	2.0
Inheritance compatibility	1.0	1.0
Approval precedent	0.5	1.0

Rationale

- Gene CDS 3033bp / cargo 4700bp (65% utilized)
- Vector tropism plus precedent target match: cns
- Both lysosomal proteins — cross-correction likely
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: lysosomal_storage
- Approval status: phase1/2

Match #2: RGX-121

Precedent disease: Mucopolysaccharidosis type II

Vector: AAV9

Tissue target: CNS/liver

Composite score: 8.5 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	1.0	2.0
Tissue tropism	2.0	2.0
Protein class	2.0	2.0
Pathway similarity	2.0	2.0
Inheritance compatibility	0.7	1.0
Approval precedent	0.8	1.0

Rationale

- Gene CDS 3033bp / cargo 4700bp (65% utilized)
- Vector tropism plus precedent target match: cns, liver
- Both lysosomal proteins — cross-correction likely
- LOF inheritance — compatible for gene replacement
- Pathway match: lysosomal_storage
- Approval status: phase3

Match #3: ST-920

Precedent disease: Fabry disease

Vector: AAV2/6

Tissue target: liver

Composite score: 7.7 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	1.0	2.0
Tissue tropism	1.5	2.0
Protein class	2.0	2.0
Pathway similarity	2.0	2.0
Inheritance compatibility	0.7	1.0
Approval precedent	0.5	1.0

Rationale

- Gene CDS 3033bp / cargo 4700bp (65% utilized)
- Precedent target match: liver
- Both lysosomal proteins — cross-correction likely
- LOF inheritance — compatible for gene replacement

- Pathway match: lysosomal_storage
- Approval status: phase1/2

Match #4: DTX201

Precedent disease: Hemophilia A

Vector: AAV8

Tissue target: liver

Composite score: 6.8 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	1.0	2.0
Tissue tropism	2.0	2.0
Protein class	2.0	2.0
Pathway similarity	0.5	2.0
Inheritance compatibility	0.7	1.0
Approval precedent	0.6	1.0

Rationale

- Gene CDS 3033bp / cargo 4700bp (65% utilized)
- Vector tropism plus precedent target match: liver
- Both secreted proteins — systemic delivery viable
- LOF inheritance — compatible for gene replacement
- Different pathway (lysosomal_storage vs coagulation)
- Approval status: phase2

Match #5: Hemgenix

Precedent disease: Hemophilia B

Vector: AAV5

Tissue target: liver

Composite score: 6.7 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	1.0	2.0
Tissue tropism	1.5	2.0
Protein class	2.0	2.0
Pathway similarity	0.5	2.0
Inheritance compatibility	0.7	1.0
Approval precedent	1.0	1.0

Rationale

- Gene CDS 3033bp / cargo 4700bp (65% utilized)

- Precedent target match: liver
- Both secreted proteins — systemic delivery viable
- LOF inheritance — compatible for gene replacement
- Different pathway (lysosomal_storage vs coagulation)
- Approval status: approved